

# FORENSICS

## 2 - Disks, Network

STT



!! nos4a2  
@sluzyp

AFSC19.EXE  
@afsc19

# ROAD MAP

~~4. Foundations~~

2. Disks, registries,  
network

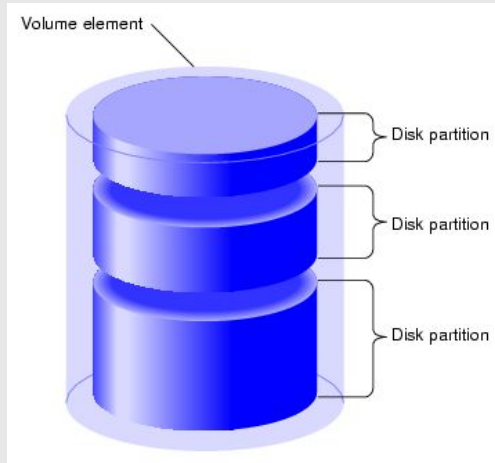
3. Memory

4. ???

# Disks/volumes



# Disk partitions



**/dev/nvme0n1 - GParted**

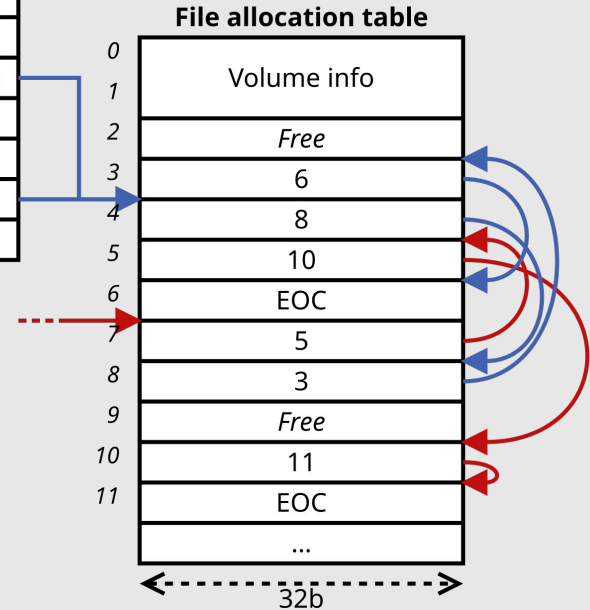
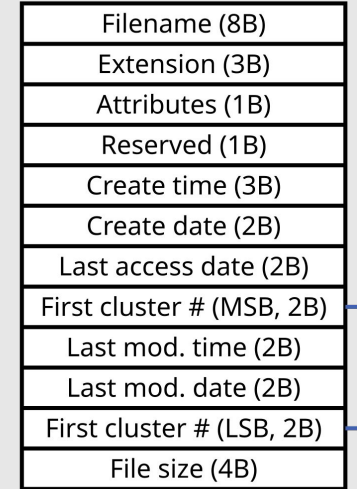
GParted Edit View Device Partition Help

/dev/nvme0n1 (476.94 GiB) ▾

| Partition      | Name                         | File System | Mount Point | Label      | Size        | Used       | Unused     | Flags        |
|----------------|------------------------------|-------------|-------------|------------|-------------|------------|------------|--------------|
| /dev/nvme0n1p1 | EFI system partition         | fat32       | /boot/efi   | SYSTEM_DRV | 260.00 MiB  | 25.16 MiB  | 234.84 MiB | boot, esp    |
| /dev/nvme0n1p2 | Microsoft reserved partition | unknown     |             |            | 16.00 MiB   | ---        | ---        | msftres      |
| /dev/nvme0n1p3 |                              | btrfs       | /           |            | 185.64 GiB  | 35.47 GiB  | 150.17 GiB |              |
| /dev/nvme0n1p4 |                              | btrfs       | /home       |            | 282.60 GiB  | 114.70 GiB | 167.90 GiB |              |
| /dev/nvme0n1p7 |                              | linux-swap  |             |            | 7.45 GiB    | 273.50 MiB | 7.18 GiB   | swap         |
| unallocated    |                              | unallocated |             |            | 1.00 MiB    | ---        | ---        |              |
| /dev/nvme0n1p5 | Basic data partition         | ntfs        |             | WINRE_DRV  | 1000.00 MiB | 395.23 MiB | 604.77 MiB | hidden, diag |

# Disk partitions

# Partitions have structures!



# Delete ≠ Destroy

Deleting data = removing the pointer from the structure

The content is still there!



Directory table entry (32B)

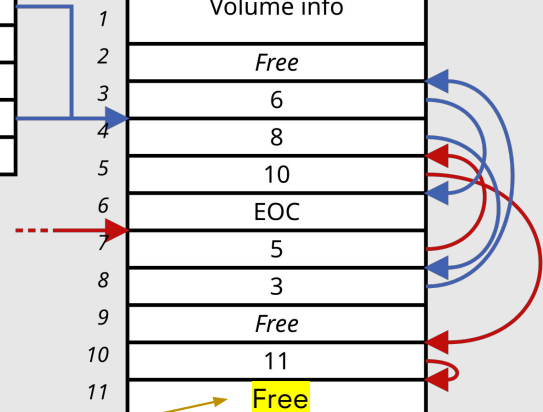
|                           |
|---------------------------|
| Filename (8B)             |
| Extension (3B)            |
| Attributes (1B)           |
| Reserved (1B)             |
| Create time (3B)          |
| Create date (2B)          |
| Last access date (2B)     |
| First cluster # (MSB, 2B) |
| Last mod. time (2B)       |
| Last mod. date (2B)       |
| First cluster # (LSB, 2B) |
| File size (4B)            |

File allocation table

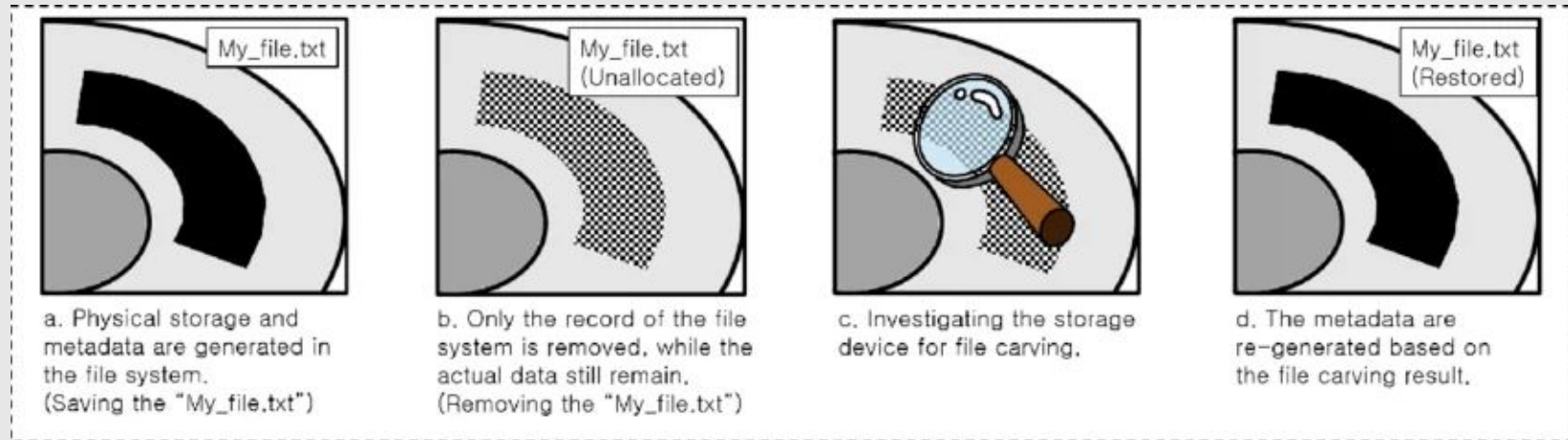
|    |             |
|----|-------------|
| 0  | Volume info |
| 1  |             |
| 2  | Free        |
| 3  | 6           |
| 4  | 8           |
| 5  | 10          |
| 6  | EOC         |
| 7  | 5           |
| 8  | 3           |
| 9  | Free        |
| 10 | 11          |
| 11 | Free        |
|    | ...         |

←----- 32b ----->

Mark as free



# We can recover deleted files! (suppostly™)



# Recovering files

## Tools:

- Testdisk
- Sleuthdisk
- Autopsy
- FTK Imager
- ... (and a lot more)

## Carving files also works:

- Binwalk
- Foremost





# Recovering files - The Sleuth Kit (TSK)

## Check partitions (mmls)

```
➤ mmls dds1-alpine.flag.img
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

   Slot      Start      End      Length    Description
000: Meta    0000000000    0000000000    0000000001 Primary Table (#0)
001: -----  0000000000    0000002047    0000002048 Unallocated
002: 000:000  0000002048    0000262143    0000260096 Linux (0x83)
```

## Information (istat + fsstat)

```
➤ istat -o 2048 dds1-alpine.flag.img 12
inode: 12
Allocated
Group: 0
Generation Id: 1464366224
uid / gid: 0 / 0
mode: rwxr-xr-x
size: 0
num of links: 1
Inode Times:
(...)
Direct Blocks:
(...)
```

## List (fls -o <off> disk <inode>)

```
➤ fls -o 2048 dds1-alpine.flag.img
d/d 10161:  home
d/d 11:    lost+found
r/r 12:    .dockerenv
d/d 2033:  bin
d/d 8129:  boot
d/d 26418: root
...
```

## Cat (icat -o <off> disk <inode>)

```
~/Downloads/reun
➤ icat -o 2048 dds1-alpine.flag.img 12
DEBUG_CHALL=0
ZIP_KEY=suprd0p3
```

# Important file locations - Linux

Always check folders you would use!

- /home/<user>/Downloads
- /home/<user>/Documents
- /root
- /tmp
- /etc (for configs)
- /var/log (for logs)

Cool files

- /home/<user>/.bash\_history
- /home/<user>/.<shell>\_\*  
(like .zsh\_rc)
- ^ The same inside /root!

# Important file locations - Windows

Always check folders you would use!

- C:\Users\<user>\Downloads
- C:\Users\<user>\Desktop
- C:\Users\<user>\AppData\\*\temp
- C:\temp
- C:\ProgramData

Cool files

- C:\\$MFT (Master File Table)
- C:\System32\config\\*  
(Registry Hives)
- C:\Users\<user>\NTUSER.DAT
- C:\Windows\System32\Winevt\Logs\\*
-

# Important file locations - Browsers

Check browsers folders for cookies, history and downloads!

Linux:

- `~/.config/<browser>/Default`
- `~/.mozilla/firefox/Default`
- `~/snap/<browser>/common/Default` (installed with snap)
- `~/var/app/<browser>/Default` (installed with flatpak)

Windows:

- `%LocalAppData%\Google\Chrome\User Data\Default\`
- `%LocalAppData%\Microsoft\Edge\User Data\Default\`
- `%AppData%\Mozilla\Firefox\Profiles\*`

Open the .db's or use NIRSOFr's (or lookup) parsing tools

# Network - packets

Semiblock



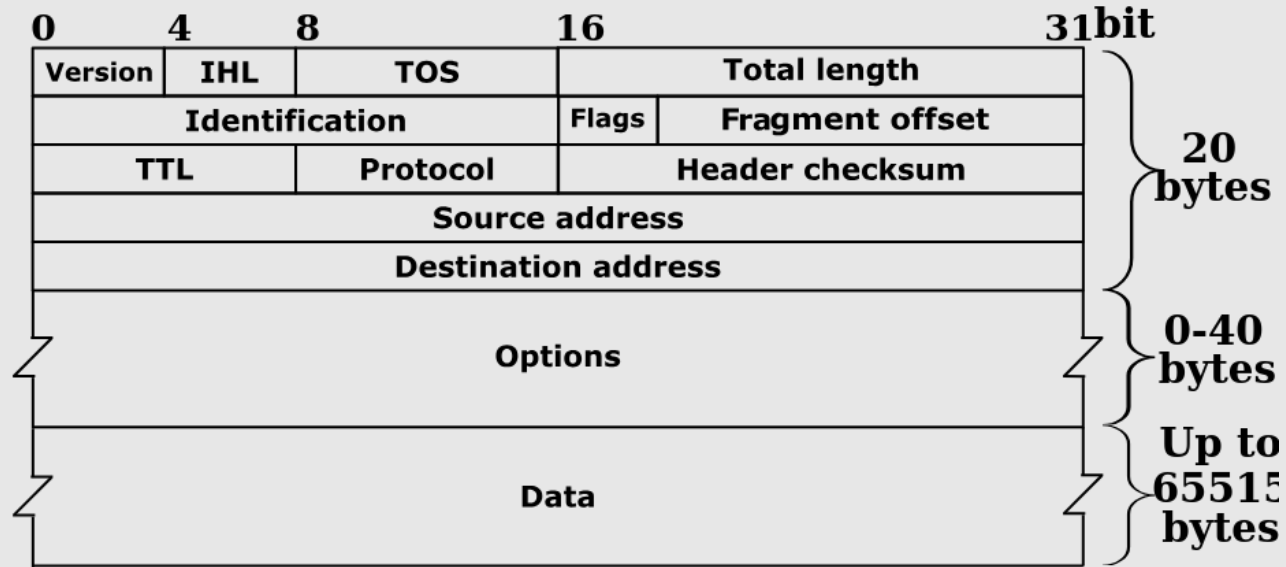
Modified block



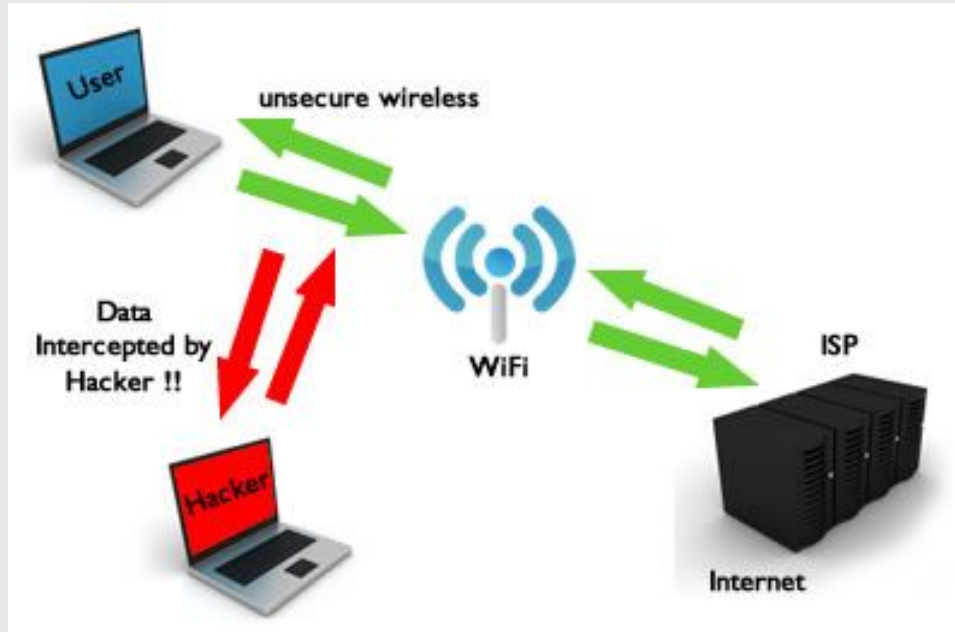
Block



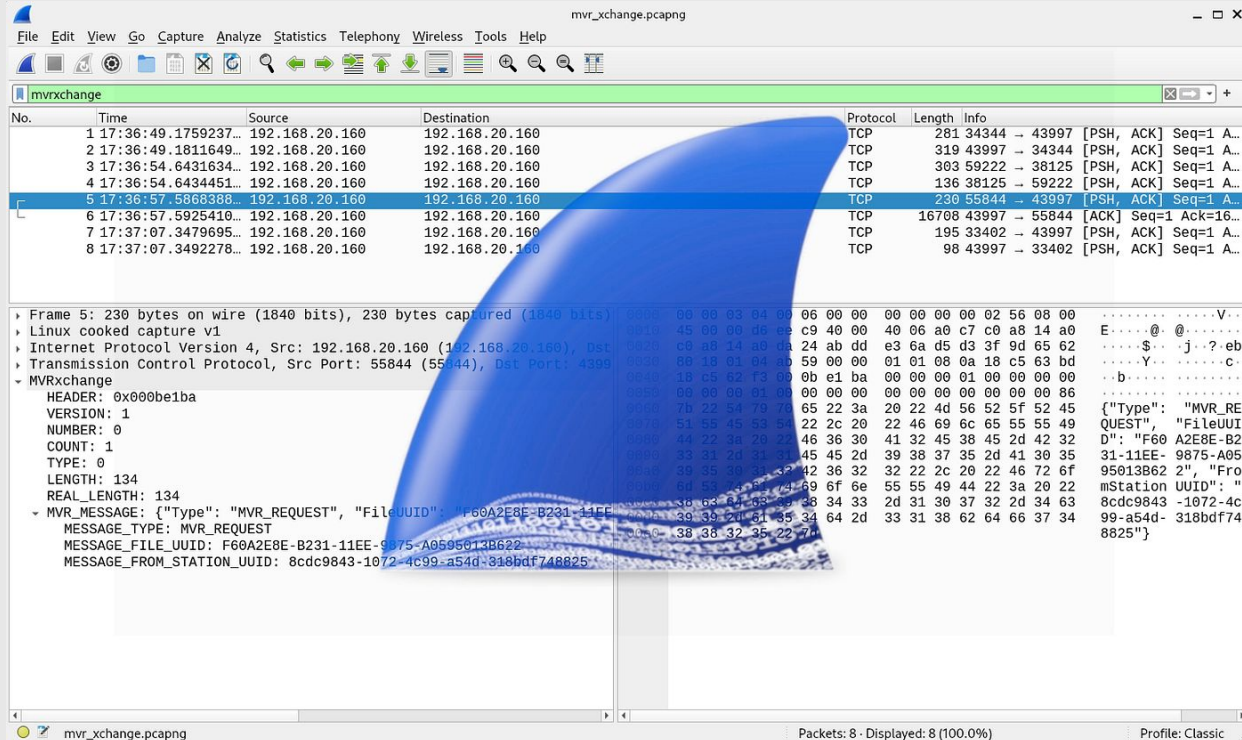
# Network - packet structures



# Network - We can sniff packets! (Wifi or router)



# Network - Analyse packets - Wireshark



The image shows a Wireshark network packet capture analysis of a file named `mvr_xchange.pcapng`. The interface displays a list of captured packets in the top pane, with packet 5 selected. The bottom pane shows the detailed view of this packet, which is a TCP Reset (RST) segment.

**Packet List:**

| No. | Time                | Source         | Destination    | Protocol | Length | Info                               |
|-----|---------------------|----------------|----------------|----------|--------|------------------------------------|
| 1   | 17:36:49.1759237... | 192.168.20.160 | 192.168.20.160 | TCP      | 281    | 34344 → 43997 [PSH, ACK] Seq=1 A.. |
| 2   | 17:36:49.1811649... | 192.168.20.160 | 192.168.20.160 | TCP      | 319    | 43997 → 34344 [PSH, ACK] Seq=1 A.. |
| 3   | 17:36:54.6431634... | 192.168.20.160 | 192.168.20.160 | TCP      | 303    | 59222 → 38125 [PSH, ACK] Seq=1 A.. |
| 4   | 17:36:54.6434451... | 192.168.20.160 | 192.168.20.160 | TCP      | 136    | 38125 → 59222 [PSH, ACK] Seq=1 A.. |
| 5   | 17:36:57.5868388... | 192.168.20.160 | 192.168.20.160 | TCP      | 230    | 55844 → 43997 [PSH, ACK] Seq=1 A.. |
| 6   | 17:36:57.5925410... | 192.168.20.160 | 192.168.20.160 | TCP      | 16708  | 43997 → 55844 [ACK] Seq=1 Ack=16.. |
| 7   | 17:37:07.3479695... | 192.168.20.160 | 192.168.20.160 | TCP      | 195    | 33402 → 43997 [PSH, ACK] Seq=1 A.. |
| 8   | 17:37:07.3492278... | 192.168.20.160 | 192.168.20.160 | TCP      | 98     | 43997 → 33402 [PSH, ACK] Seq=1 A.. |

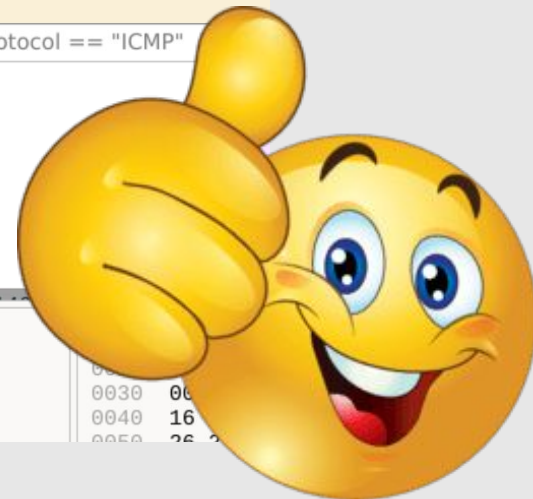
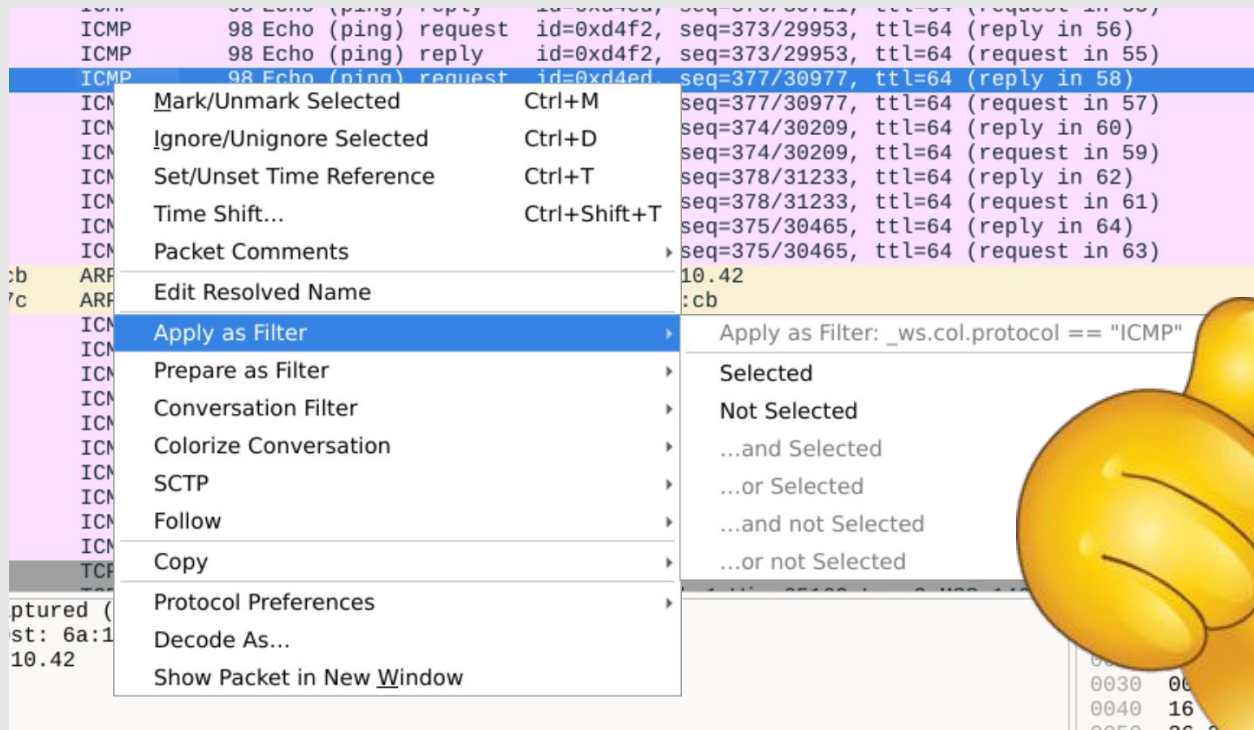
**Packet 5 Details:**

- Frame 5: 230 bytes on wire (1840 bits), 230 bytes captured (1840 bits) on interface
- Linux cooked capture v1
- Internet Protocol Version 4, Src: 192.168.20.160, Dst: 192.168.20.160
- Transmission Control Protocol, Src Port: 55844 (55844), Dst Port: 43997
- MVRxchange
  - HEADER: 0x000be1ba
  - VERSION: 1
  - NUMBER: 0
  - COUNT: 1
  - TYPE: 0
  - LENGTH: 134
  - REAL\_LENGTH: 134
  - MVR\_MESSAGE: {"Type": "MVR\_REQUEST", "FileUUID": "F60A2E8E-B231-11EE-9075-A0595613B622", "Message": "MVR\_REQUEST", "Message\_File\_UUID: F60A2E8E-B231-11EE-9075-A0595613B622", "Message\_From\_Station\_UUID: 8cdc9843-1072-4c99-a54d-318bdf748825"}

The status bar at the bottom indicates: `mvr_xchange.pcapng`, Packets: 8 - Displayed: 8 (100.0%), Profile: Classic.



# Wireshark - use filters!



# Practice!

- Solve challenges on CMU's Cylab (picoCTF) - [learn.cylabacademy.org](https://learn.cylabacademy.org)
- Try Sherlocks - [hackthebox.com](https://hackthebox.com) (PS: root-me + Cyberdefenders)

